

FINAL YEAR PROJECT PROPOSAL

Project Title

AI Powered Personalized Academic and Career Decision Support System

1. Introduction

Choosing the right degree program and university is an important decision for students. However, students often have to search through multiple sources to find information about university recognition, available programs, eligibility criteria, tuition fees, scholarships, admission requirements, career opportunities, and required skills. This information is usually distributed across Higher Education Commission resources, university websites, prospectuses, and other official documents.

The proposed project aims to develop an AI powered academic and career decision support system that brings this information together and provides personalized recommendations based on an individual student's academic background, interests, career goals, budget, location, and preferences.

Unlike a conventional university search website, the proposed system will not simply display a list of universities. It will analyze the student's profile, determine eligibility, identify suitable academic programs, rank universities, explain the reasons behind recommendations, and provide a possible career pathway associated with the selected program.

The system will combine Natural Language Processing, semantic embeddings, machine learning, Knowledge Graphs, Retrieval Augmented Generation, Explainable AI, and Multi Agent AI to provide personalized and evidence based academic guidance.

2. Background

Students often face difficulty when selecting a university or degree because academic information is distributed across different platforms. A student may need to visit several university websites to compare programs, eligibility requirements, fees, scholarships, admission requirements, and career opportunities.

Traditional university search systems generally work through keyword based searches and filters. These systems do not deeply understand the student's interests, career objectives, academic strengths, or financial constraints.

For example, a student may state:

"I have 82 percent in ICS, I enjoy programming and mathematics, I want to become an AI Engineer, my budget is Rs. 150,000 per year, and I prefer Islamabad."

A conventional search engine may only search for universities offering Artificial Intelligence programs. The proposed system will analyze the complete student profile and determine which programs and universities are most suitable for that particular student.

3. Problem Statement

Students currently lack a centralized intelligent system that can integrate academic, institutional, financial, and career information and use it to provide personalized and explainable recommendations.

Existing systems generally focus on searching or filtering universities and programs. They do not adequately combine student interests, academic performance, career goals, budget, eligibility, program content, and career pathways into one decision making process.

Furthermore, university information such as fees, admission requirements, scholarships, and available programs can change over time. If outdated information is used, students may receive incorrect recommendations.

Therefore, there is a need for an intelligent system that can provide personalized recommendations using verified information while also explaining the reasoning and evidence behind each recommendation.

4. Research Question

How can heterogeneous academic, institutional, and career information be integrated with machine learning, semantic computing, knowledge graphs, and Retrieval Augmented Generation to provide personalized, explainable, and trustworthy academic and career recommendations?

5. Aim of the Project

The main aim of this project is to develop an AI powered academic and career decision support system that helps students identify suitable degree programs and universities based on their academic profile, interests, career goals, financial constraints, location preferences, and eligibility.

The system will also provide evidence based explanations, career pathways, skill gap analysis, and what if analysis to support better academic decision making.

6. Objectives

The objectives of the proposed system are:

1. To develop a student profiling system that collects academic, personal preference, interest, career, budget, and location information.
2. To use Natural Language Processing to understand students' natural language descriptions of their interests and career goals.

3. To use semantic embeddings to identify similarities between student interests and academic programs.
4. To develop an eligibility checking mechanism based on university and program admission requirements.
5. To develop a machine learning based recommendation and ranking model using XGBoost.
6. To construct a Knowledge Graph connecting universities, academic programs, courses, skills, and career opportunities.
7. To implement a Retrieval Augmented Generation system for retrieving reliable information from verified academic documents.
8. To provide explanations for recommendations using Explainable AI techniques such as SHAP.
9. To implement specialized AI agents for student analysis, eligibility checking, program recommendation, university research, career planning, and explanation.
10. To develop a data freshness mechanism that tracks the source and verification date of university information.
11. To provide what if analysis so students can observe how changes in budget, location, or other preferences affect recommendations.
12. To evaluate the recommendation system using appropriate machine learning and information retrieval metrics.

7. Proposed Solution

The proposed system will work as an intelligent academic counselor.

A student will provide information such as academic marks, educational background, subjects, interests, career goals, budget, preferred location, and other preferences.

The system will first process the student's information and create a structured student profile.

The profile will then be analyzed through several computational components.

Natural Language Processing will extract important information from natural language input.

Semantic embeddings will convert student interests and program descriptions into numerical representations and calculate semantic similarity.

The eligibility engine will determine whether the student meets the admission requirements of different programs.

Eligible programs will be passed to the recommendation model. XGBoost will analyze multiple features and generate suitability scores for candidate universities and programs.

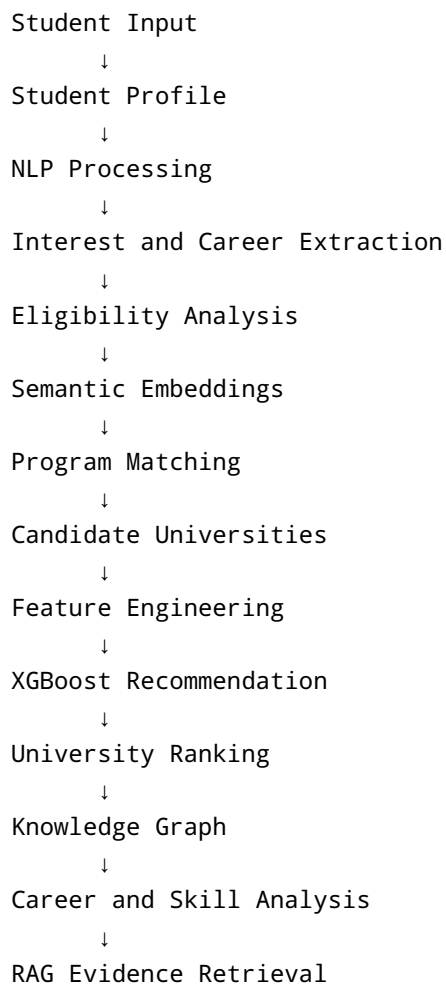
The Knowledge Graph will connect academic programs with courses, skills, and careers.

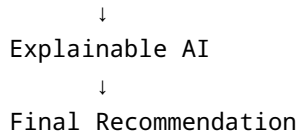
The RAG system will retrieve relevant information from verified university documents and official sources whenever the student asks questions or requires evidence.

Finally, the system will generate a personalized recommendation with an explanation of why each option was selected.

8. System Workflow

The overall workflow of the proposed system will be:





9. Major Features

9.1 Student Profile

The system will allow students to enter academic information, interests, career goals, budget, preferred location, and other preferences.

9.2 Natural Language Input

Students will be able to describe their interests and goals using normal language instead of selecting only predefined options.

For example:

"I like mathematics and programming and want to work in Artificial Intelligence."

The system will extract relevant interests and career information.

9.3 Eligibility Checker

The system will compare the student's academic profile with the admission requirements of different programs and identify eligible and ineligible options.

9.4 Personalized Program Recommendation

The system will recommend academic programs based on the student's interests, academic background, career objectives, and semantic similarity with program descriptions.

9.5 University Ranking

Eligible universities will be ranked according to multiple factors including academic compatibility, interest match, career compatibility, budget compatibility, location preference, and other relevant features.

9.6 University Comparison

Students will be able to compare multiple universities based on factors such as fees, eligibility, program compatibility, location, scholarships, and career alignment.

9.7 Career Pathway

The system will show a possible pathway from an academic program to relevant skills, job roles, and career opportunities.

9.8 Skill Gap Analysis

The system will compare the student's existing skills with the skills required for a selected career and identify missing skills.

9.9 Evidence Based Question Answering

Students will be able to ask questions about universities and programs. The RAG system will retrieve relevant information from verified documents before generating an answer.

9.10 Explainable Recommendations

The system will explain why a particular university or program received a high recommendation score.

9.11 What If Analysis

Students will be able to change constraints such as budget or location and observe how their recommendations change.

9.12 Data Freshness

The system will maintain information about the source and last verification date of university data.

10. Artificial Intelligence and Algorithms

10.1 Natural Language Processing

Natural Language Processing will be used to understand natural language provided by students.

It will identify information such as interests, career goals, skills, and preferences.

10.2 Semantic Embeddings

Sentence Transformers will convert student descriptions and university program descriptions into vector representations.

This will allow the system to compare meanings rather than relying only on exact keywords.

10.3 Cosine Similarity

Cosine Similarity will be used to calculate the semantic similarity between student interests and academic program descriptions.

A higher similarity score will indicate stronger alignment between the student's interests and a particular program.

10.4 XGBoost

XGBoost will be used as the primary supervised machine learning algorithm for university and program suitability ranking.

The model will use features such as academic compatibility, semantic interest match, career match, budget compatibility, location preference, and eligibility.

The output will be a suitability score that can be used to rank candidate universities.

10.5 Knowledge Graph

A Knowledge Graph will represent relationships between universities, programs, courses, skills, and careers.

For example:

```
graph TD; A[BS Artificial Intelligence] --> B[Machine Learning]; B --> C[Deep Learning]; C --> D[Natural Language Processing]; D --> E[AI Engineer];
```

BS Artificial Intelligence
↓
Machine Learning
↓
Deep Learning
↓
Natural Language Processing
↓
AI Engineer

This will support career pathway and skill relationship analysis.

10.6 Vector Search

Vector search will retrieve documents that are semantically similar to a student's question.

It will allow the system to find relevant information even when the student's wording does not exactly match the wording in the source document.

10.7 Retrieval Augmented Generation

RAG will retrieve relevant information from the verified knowledge base before providing an answer through the language model.

This approach will reduce dependence on the language model's internal knowledge and improve the reliability of university related answers.

10.8 Reranking

A reranking mechanism will be used after initial document retrieval to identify the most relevant documents or document sections for a particular question.

10.9 SHAP

SHAP will be used to explain the contribution of different features to the machine learning recommendation.

For example, the system may explain that a university received a high score because of strong career alignment, academic compatibility, interest similarity, and budget compatibility.

11. Data Sources

The system will prioritize authoritative sources.

Higher Education Commission

HEC information will be used for university recognition, institutional information, and relevant qualification information.

Pakistan Qualification Register

The Pakistan Qualification Register will be used where applicable for recognized qualifications and academic information.

Official University Websites

Official university websites and official documents will be used for:

Program information

Eligibility criteria

Fee structures

Admission requirements

Scholarships

Course information

Prospectuses

Admission deadlines

The system will store source information and verification dates along with the collected data.

12. Data Engineering and Freshness Pipeline

Because academic information can change, the project will include a data update mechanism.

The proposed pipeline will be:

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graph TD; A[Official Sources] --> B[Scheduled Data Collection]; B --> C[Data Cleaning]; C --> D[Normalization]; D --> E[Duplicate Removal]; E --> F[Validation]; F --> G[Change Detection]; G --> H[Database Update]; H --> I[Vector Knowledge Base Update];
```

The system will not claim to provide 100 percent real time information. Instead, it will provide near real time automated synchronization based on the availability of updated information from official sources.

Each important data item will include its source and last verification date.

13. Multi Agent AI Architecture

The system will use specialized agents for different responsibilities.

Student Analysis Agent

Understands the student's academic profile, interests, preferences, and career goals.

Eligibility Agent

Checks whether the student satisfies program admission requirements.

Program Agent

Identifies academic programs that align with the student's profile.

University Agent

Analyzes and ranks suitable universities.

Career Agent

Develops career pathways and identifies relevant skills.

Research Agent

Retrieves information from the knowledge base and university documents.

Explanation Agent

Generates understandable explanations for recommendations.

An orchestrator will coordinate these specialized components.

14. Explainable AI

Trust is important because the system will influence educational decisions.

Therefore, the system will not simply provide:

University A: 94 percent

Instead, it will provide an explanation such as:

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Career Match: 25 percent
Interest Match: 22 percent
Academic Match: 20 percent
Budget Compatibility: 15 percent
Location Match: 8 percent
Other Factors: 4 percent
```

The explanation will help the student understand the factors influencing the recommendation.

15. What If Analysis

The system will support decision simulation.

For example, if a student's budget is Rs. 150,000, the system may rank universities according to that constraint.

If the student changes the budget to Rs. 250,000, the system will recalculate the relevant features and generate a new ranking.

The same approach can be applied to other preferences such as location, program preference, or admission requirements.

This makes the system a decision support tool rather than a static recommendation website.

16. Proposed Technology Stack

Frontend

React

TypeScript

Tailwind CSS

Backend

Python

FastAPI

Database

PostgreSQL

Vector Database

FAISS or Qdrant

Knowledge Graph

Neo4j

Machine Learning

XGBoost

Scikit Learn

NLP

Sentence Transformers

Embedding Models

Large Language Model

RAG

Vector Retrieval

Reranking

Document Processing

Large Language Model

Explainable AI

SHAP

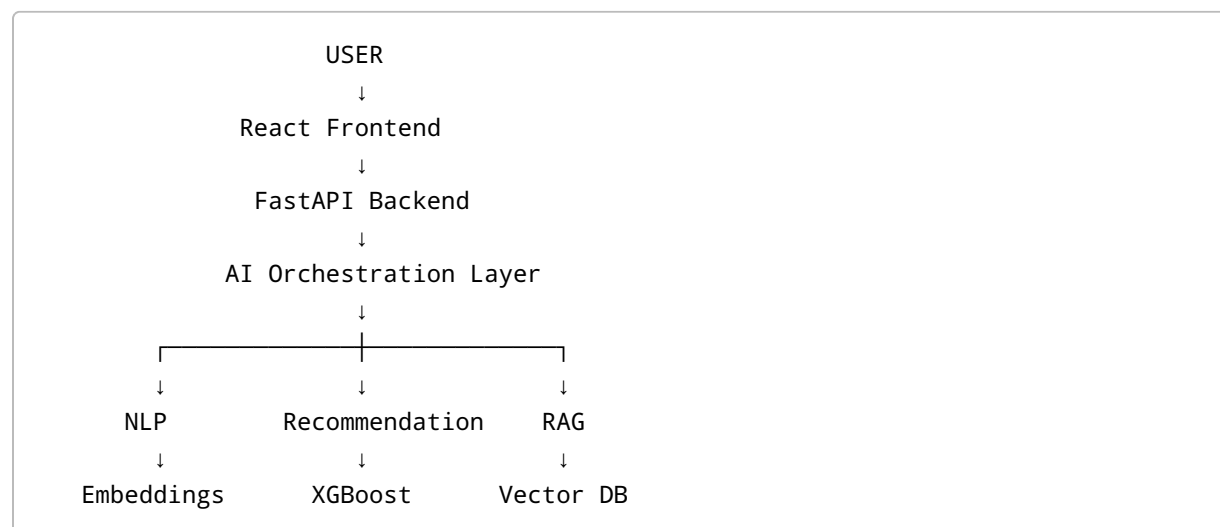
Data Processing

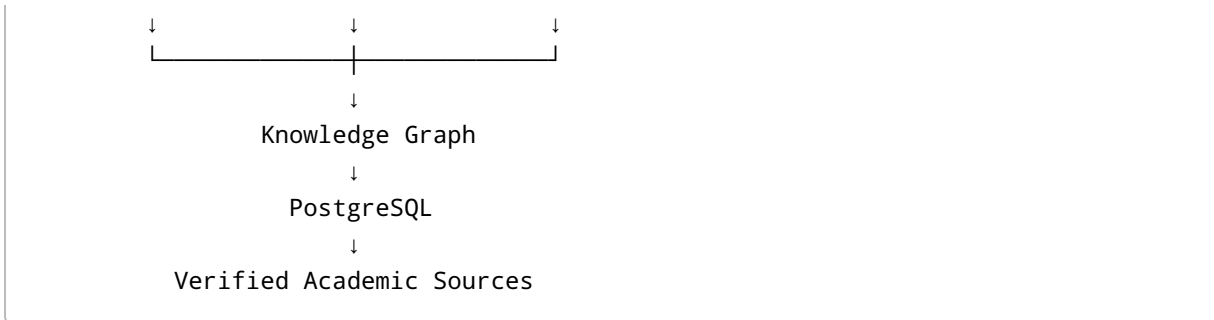
Python

Data Cleaning and Validation Libraries

17. System Architecture

The proposed architecture will consist of several layers.





18. Methodology

The project will follow an Agile based development methodology combined with an AI and machine learning development process.

Phase 1: Requirement Analysis

The requirements of students, universities, and system administrators will be identified.

Phase 2: Data Collection

Academic and institutional information will be collected from HEC, PQR, official university websites, and official documents.

Phase 3: Data Engineering

Collected information will be cleaned, normalized, validated, and stored in structured and unstructured formats.

Phase 4: Knowledge Base Development

PostgreSQL will store structured information, a vector database will store embeddings, and Neo4j will store relationships between academic and career entities.

Phase 5: AI Model Development

Semantic embedding and similarity methods will be implemented for matching. XGBoost will be developed for recommendation and ranking.

Phase 6: RAG Development

Academic documents will be processed, embedded, indexed, retrieved, reranked, and provided to the language model for evidence based responses.

Phase 7: Multi Agent Development

Specialized agents will be developed and connected through an orchestration mechanism.

Phase 8: System Integration

The AI components will be integrated with the FastAPI backend and React frontend.

Phase 9: Evaluation

The system will be evaluated using recommendation, retrieval, response quality, performance, and user satisfaction metrics.

19. Evaluation

The recommendation system will be evaluated using appropriate metrics.

Precision at K

Measures how many of the top K recommendations are relevant.

Recall at K

Measures how many relevant options are successfully retrieved within the top K recommendations.

NDCG

Measures the quality of the ranking while giving higher importance to highly relevant results appearing near the top.

RAG Faithfulness

Measures whether generated answers are supported by the retrieved evidence.

Retrieval Accuracy

Measures whether the system retrieves the correct or relevant source information.

Response Time

Measures how quickly the system produces recommendations and answers.

Data Freshness

Measures how recently important university information has been verified.

User Satisfaction

Students can evaluate the usefulness, clarity, relevance, and trustworthiness of recommendations.

20. Innovation and Novelty

The main innovation of this project is the integration of multiple AI and computational techniques into a single academic decision support system.

The proposed system combines personalized student profiling, semantic matching, machine learning based ranking, Knowledge Graph reasoning, evidence based RAG, explainable recommendations, career pathway generation, skill gap analysis, and what if analysis.

Rather than simply answering university related questions, the system will support the complete academic decision making process.

The project will also emphasize trustworthy information by linking recommendations and answers to verified academic sources and maintaining data freshness information.

21. Scope

The initial scope of the project will focus on universities and academic programs for which reliable official information can be collected.

The system will primarily support undergraduate academic decision making.

The system will consider factors such as academic eligibility, interests, career goals, budget, location, program compatibility, and available university information.

The project will provide decision support and recommendations rather than making an irreversible decision on behalf of the student.

22. Limitations

The accuracy of the system will depend on the quality and availability of official university information.

The system cannot guarantee that information is current if a university has not published a recent update.

Career predictions cannot guarantee a specific employment outcome because career success depends on many factors beyond academic programs.

The recommendation score should be treated as a decision support indicator rather than an absolute measure of university quality.

23. Expected Outcomes

At the completion of the project, the system is expected to provide:

1. A personalized student profiling system.

2. An eligibility checking mechanism.
3. AI based academic program matching.
4. Machine learning based university ranking.
5. A Knowledge Graph connecting programs, skills, and careers.
6. A RAG based academic question answering system.
7. Evidence based university information.
8. Explainable recommendations.
9. Career pathway generation.
10. Skill gap analysis.
11. What if decision simulation.
12. University data freshness and source tracking.
13. A user friendly web based interface.

24. Significance of the Project

The proposed system can help students reduce the time and effort required to research academic opportunities.

Instead of visiting multiple websites and manually comparing information, students will be able to provide their requirements and receive personalized recommendations.

The system can also improve transparency by explaining why a recommendation was generated and by providing supporting evidence from authoritative sources.

From a technical perspective, the project demonstrates the practical integration of machine learning, natural language processing, semantic computing, Knowledge Graphs, information retrieval, RAG, multi agent AI, and explainable AI in a real world decision support problem.

25. Conclusion

The proposed project is an AI powered academic and career decision support system designed to help students make informed decisions about their education and future careers.

The system will analyze student characteristics, academic performance, interests, career goals, financial constraints, and preferences. It will combine eligibility rules, semantic embeddings, XGBoost based ranking, Knowledge Graph reasoning, RAG, multi agent AI, and Explainable AI to generate personalized and evidence based recommendations.

The project will go beyond traditional university search systems by answering not only where a student can study, but also what they should study, whether they qualify, why a particular option is recommended, what skills they need, and what career pathway they can follow.

The final system will therefore function as an intelligent academic and career counselor that supports students throughout the decision making process.

26. One Line Project Definition

An AI powered system that tells students what to study, where to study, whether they qualify, why the option is recommended, what career path it leads to, and what evidence supports the decision.